PNAS RESEARCH ARTICLE | Collective behavior

From hunter to shepherd: does the optimal alignment of a fish school depend on the agent that perturbs it?

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Fish schools under attack are thought to benefit from being close to the transition between disordered and ordered motion. Klamser and Romanczuk showed with a predator–prey model that schools are indeed caught least there. According to them, the reason is mainly the looser structure of the school, not a better collective response. This project asks whether this optimum still holds when the school faces a different agent: first a more realistic predator, then a shepherd that tries to move the school instead of catching prey. In this first report, models of collective motion, predation and shepherding are reviewed. The model of Klamser and Romanczuk is reimplemented in Python and validated against the original code, and the minimum of the capture rate at the transition is reproduced (about 13% fewer captures). The next reports will test whether this minimum stays at the transition for a predator that is confused by many prey or that intercepts its target, in 2D and 3D, and whether the transition also helps a shepherd.

Introduction

When a predator attacks a fish school, the prey do not react independently. They turn, split and regroup together, so the predator has to deal with a moving target that keeps changing shape. One explanation of how such groups process information is the criticality hypothesis: living groups would operate close to a phase transition, where their response to perturbations is largest (1, 2). For moving groups, the transition of interest is the one between a disordered swarm and an ordered, polarized school. It appears in models of self-propelled particles when alignment becomes stronger than noise (3, 4).

Klamser and Romanczuk (5) tested this hypothesis with a model in which a single predator attacks a school of 400 prey. The capture rate is lowest near the order–disorder transition, but not for the reason the hypothesis gives. They compared the attacked school with an identical school that does not flee. The fraction of prey saved by fleeing keeps increasing with alignment and has no peak at the transition. The advantage of the critical school therefore comes from its spatial structure, which is looser, and not from a better collective response. They also let the alignment strength evolve. Each prey survives better by aligning more, so the population ends up far in the ordered phase: the transition is best for the group, but evolution does not keep it there.

These results are obtained with one specific predator. It swims twice as fast as the prey, heads for a weighted center of the prey in front of it, and always follows the same rule. Real predators choose the direction of their attack (marlins most often attack sardines from the side (6)), combine tactics such as splitting a school before isolating a prey (7), and get confused when many prey move around them (8). Would a different predator change which alignment protects a school best? The opposite question is also open. The high responsiveness at criticality does not explain the low capture rate, but it could be useful for an agent that wants to steer a school without breaking it, like a sheepdog with a flock (9) or a group of robots with fish, to lead them away from dangers or along a given path (10, 11).

The question is therefore whether the benefit of being near the order–disorder transition depends on the agent that perturbs the school: does the transition still protect the school from a more realistic predator, and does it make the school easier for a shepherd to move? The project has three steps: reproduce the model of Klamser and Romanczuk, make the predator confused by many prey or able to intercept its prey, in 2D and then 3D, then turn the predator into a shepherd that moves the school without breaking it. This first report reviews existing models, presents the reimplementation and the reproduced result, and describes the next steps.

Related work

Collective motion and criticality. Vicsek et al. (3) showed that self-propelled particles that align with their neighbors have a phase transition. Below a critical noise level the group moves in a common direction; above it, the motion is disordered.

Couzin et al. (4) added short-range repulsion and long-range attraction, and obtained the main collective states seen in nature: swarm, milling and parallel motion. Statistical models fitted to real data, from neurons to bird flocks, seem to be close to a critical point. This led to the hypothesis that biological systems are tuned to criticality (1). Muñoz (2) reviews the arguments for it, mainly a maximal sensitivity to stimuli and a large range of possible behaviors, and notes that the evidence is still indirect.

Predator–prey models. Klamser and Romanczuk (5) combine a Vicsek-like school with a pursuing predator, and evolve the alignment strength of the prey with an evolutionary algorithm. Other studies focus on the predator itself. Demšar et al. (7) evolved composite hunting tactics with a genetic algorithm, for example splitting a school before catching an isolated prey. The dispersing tactic worked much better when the predator could be confused by the prey. Olson et al. (8) showed that predator confusion alone is enough for swarming to evolve in prey. Bartashevich et al. (6), from the same group as Klamser, filmed striped marlin attacking sardine schools. The "fountain" escape of the school depended on the direction of the attack, and an agent-based model reproduced it.

Shepherding. Steering a group is the opposite task: the agent has to move the group without breaking it. Strömbom et al. (9) proposed a single shepherd that collects the agents when they are too spread out and drives them once they are grouped. Their model reproduces data from sheep and sheepdogs. However, their sheep do not align with each other and stop when the dog is far away, whereas fish swim all the time in an aligned school. Van Havermaet et al. use several robots that surround the group, to keep a herd away from dangers (10) or to guide simulated fish along a path (11). Their method needs enough robots for the size of the group. Shibayama and Kawashima (12) guided small groups of 5 to 8 real fish with one virtual agent shown on a screen and trained by reinforcement learning. The guidance got worse as the group grew, and more agents did not help. It is not known whether a single agent can steer a large aligned school, or how this depends on alignment.

Methods

Model. The model of Klamser and Romanczuk (5) and its C++ implementation are followed, with the parameters of Table 1. Each of the $N = 400$ prey moves at a constant speed v_0. Its heading $\mathbf{u}_i$ is a unit vector, which turns at each time step by

$$\delta\theta_i = \frac{1}{v_0}\left(\mathbf{F}_i \cdot \mathbf{u}_i^{\perp}\,\Delta t + \sqrt{2D\,\Delta t}\;\eta_i\right), \qquad [1]$$

where $\mathbf{u}_i^{\perp}$ is the heading rotated by 90°, D is the angular noise and η_i is a standard Gaussian variable. Each prey interacts with its Voronoi neighbors $\mathcal{N}_i$:

$$\mathbf{F}_i = \frac{1}{n_i}\sum_{j\in\mathcal{N}_i}\Big[\mu_d \tanh\big(m_d(r_{ij}-r_d)\big)\,\hat{\mathbf{r}}_{ij} + \mu_{\mathrm{alg},i}\,v_0(\mathbf{u}_j-\mathbf{u}_i)\Big] + \mu_{\mathrm{flee}}\,\hat{\mathbf{e}}_i. \qquad [2]$$

The first term keeps neighbors at a distance r_d and the second one aligns the headings. The last term pushes prey i away from the predator, along the unit vector $\hat{\mathbf{e}}_i$, when the predator is one of its neighbors. n_i is a neighbor count (see Reimplementation). The individual alignment strengths $\mu_{\mathrm{alg},i}$ are drawn around a mean value μ_{alg}.

The predator is twice as fast as the prey. It appears after an equilibration time T_{eq}, at a distance of 4.5 from the school, from an almost random direction. It first heads for the center of the school. Once it is close, it heads for the prey in front of it, weighted by their chance of being caught. A prey in front of the predator, among its Voronoi neighbors, at a distance $d < r_{\mathrm{catch}}$ is caught with probability $p = \gamma_a\,\Delta t\,(r_{\mathrm{catch}} - d)/(r_{\mathrm{catch}}\,n_c)$, where n_c is the number of prey in reach. Caught prey are removed.

Reimplementation. The model was rewritten in Python. Directions are stored as unit vectors, so the code can be extended to three dimensions. Every run takes a seed and can be replayed, which is not possible with the C++ code. When the code and the equations of the paper differ, the code is followed, because it produced the published figures. The code divides the social force by about half the number of neighbors (n_i), and n_c only counts the prey within r_{catch}.

Table 1. Model parameters (Table S1 of (5)).

Parameter	Symbol	Value
Number of prey	N	400
Prey speed, preferred distance	v_0, r_d	1
Angular noise	D	0.5
Mean alignment strength (swept)	μ_{alg}	0–1.6
Distance regulation strength, slope	μ_d, m_d	2
Flee strength	μ_{flee}	4
Predator speed, pursuit strength	v_p, μ_p	2
Catch radius	r_{catch}	3
Attack rate	γ_a	1/3
Equilibration, attack duration	$T_{\mathrm{eq}}, T_{\mathrm{simu}}$	200, 120
Time step	Δt	0.02

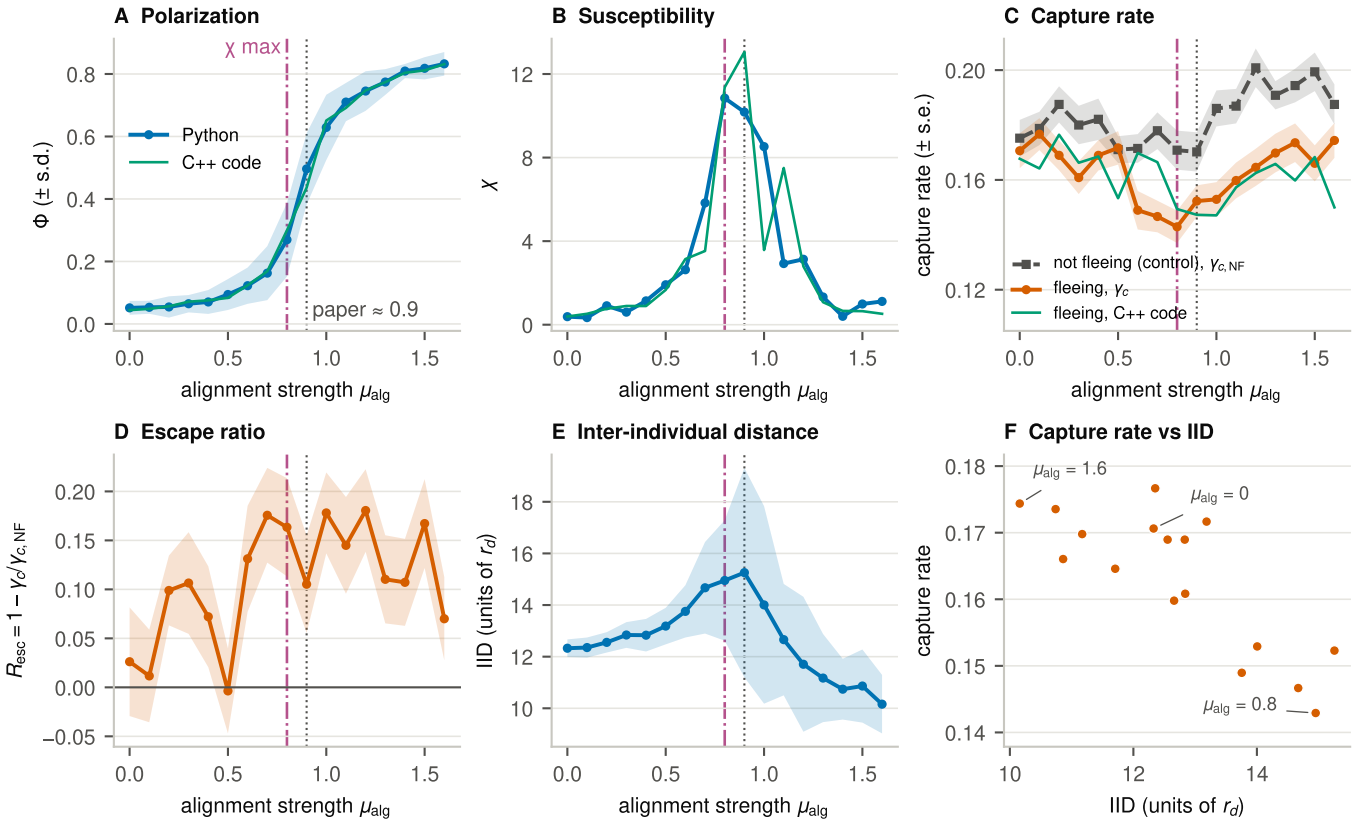


Fig. 1. Order transition and predation against the alignment strength μ_{alg} (parameters of Table 1, $D = 0.5$, 40 runs per point). (A) Polarization when the predator appears (± s.d.). (B) Susceptibility. (C) Capture rate of the fleeing school γ_c and of the control school that does not flee $\gamma_{c,\mathrm{NF}}$ (± s.e.; the y-axis does not start at 0). (D) Escape ratio (± bootstrap s.e.). (E) Inter-individual distance when the predator appears (± s.d.). (F) Capture rate against the IID, one point per alignment strength ($R = -0.80$). Shaded bands: s.d. or s.e., as given for each panel. Green lines in A–C: original C++ code, same parameters and number of runs. Dash-dotted line: maximum of χ; dotted line: transition given by the paper. Phases compared in the text: disordered ($\mu_{\mathrm{alg}} = 0$–0.2), transition (0.8–1.0), ordered (1.4–1.6).

Measures. Two quantities are measured when the predator appears: the polarization $\Phi = |\frac{1}{N}\sum_i \mathbf{u}_i|$, which is close to 0 for a disordered school and to 1 for an aligned one, and the inter-individual distance (IID), the mean distance between all pairs of prey. The transition is placed at the maximum of the susceptibility $\chi = \beta N(\langle\Phi^2\rangle - \langle\Phi\rangle^2)$, with $\beta = 1/(Dv_0)$. χ measures how much Φ varies between runs (5). The capture rate γ_c is the number of captures per unit time. Each attack is repeated on a control school that starts from the same state but does not flee. This gives a second capture rate $\gamma_{c,\mathrm{NF}}$ and the escape ratio $R_{\mathrm{esc}} = 1 - \gamma_c/\gamma_{c,\mathrm{NF}}$. The mean alignment strength μ_{alg} is varied from 0 to 1.6 in steps of 0.1, with 40 runs per value. For validation, the C++ code is run with the same values.

Planned extensions. Each extension only changes the agent that perturbs the school, and this agent only uses local information. Apart from the move to 3D, the prey rules and the sweep over μ_{alg} stay the same, and the new predators are also tested on the control school, so each new agent can be compared directly with the reproduced model.

For the second report, the predator will be changed in two ways, each aimed at one of the two effects found in the reproduction. First, it will be confused by many prey, as in the models of Olson et al. (8) and Demšar et al. (7): its chance of making a catch will decrease when many prey are within a confusion radius, counted among all the prey it sees and not only its Voronoi neighbors. In the original model this chance does not depend on their number: dividing by n_c only shares it among them. On the ordered side the dip comes from the looser structure of the school, and dense schools are protected against a confused predator. Since the ordered phase is the densest (Fig. 1E), the minimum is expected to move towards it. Second, the predator will head for the predicted position of the prey it hunts instead of their current position. On the disordered side the dip comes from fleeing, and a predator that intercepts fleeing prey should reduce this benefit. The capture rate and the escape ratio will be measured against μ_{alg}, with the original predator as reference. Each change is first tested alone, then both together. The comparison will then be made in three dimensions, with 3D Voronoi neighbors. There, the predator will also choose the direction of its attack, including from below or above, since the escape of a school depends on the attack direction (6).

For the final report, the predator becomes a shepherd that moves the school to a target region. The rules of Strömbom et al. (9) (collect the group when it is spread out, drive it when it is grouped) will be adapted to fish, with a shepherd that only sees the fish near it. The main change is that the school keeps swimming and aligning by itself, so the shepherd has to deflect a moving school, without making it flee and split. The fraction of the school that reaches the target, the time needed and the number of groups the school splits into will be measured as a function of alignment.

The shepherd is expected to work best near the transition, where the school responds strongly but stays in one group. One shepherd will also be compared with the robot groups of Van Havermaet et al. (11).

Results

Validation against the original code. C++ runs cannot be replayed, so the two codes are compared on average. The polarization curves overlap, and the susceptibility peaks one step apart. The mean capture rate is also the same: 0.161 for the C++ code and 0.163 for the Python code. After correction for multiple tests, no alignment strength shows a significant difference (Fig. 1A–C). The only clear difference is that the dip of the C++ code is less deep on the ordered side.

Order and responsiveness. When the alignment strength increases, the school goes from a disordered swarm ($\Phi \approx 0$) to a polarized school (Fig. 1A). The polarization varies most between runs around $\mu_{\mathrm{alg}} \approx 0.8$, where the susceptibility peaks (Fig. 1B). The transition is placed there, close to the value of 0.9 given in the paper.

Group optimum. The predator catches the fewest prey at the transition (Fig. 1C), as in Fig. 2C of the paper. Around it, the capture rate is about 13% lower than in the disordered or the ordered phase, and both differences are significant.

Fleeing or structure. A school can be caught less because it escapes better, or because its structure makes it harder to attack. The control school has the same structure but does not flee, so it separates the two. On average, fleeing helps: the control school loses more prey, and the escape ratio is about 0.1. However, the escape ratio has no peak at the transition (Fig. 1D): it rises from the disordered phase, then stays roughly constant. On the ordered side, the control school is also caught least near the transition, so on this side the dip comes mostly from the structure of the school. On the disordered side, the control school is barely lower at the transition, so on this side the dip comes mostly from fleeing. Near the transition the school is also the most spread out (Fig. 1E). Across alignment strengths, the inter-individual distance is strongly anticorrelated with the capture rate ($R = -0.80$, Fig. 1F), as in the paper ($R = -0.69$).

Discussion

The first result of Klamser and Romanczuk (5) is reproduced: a school is caught least near the order–disorder transition, and, as in the paper, the escape ratio has no peak there. The dip comes mostly from the structure of the school on the ordered side, and from fleeing on the disordered side.

There are two differences from the paper, and one caveat. First, the capture rates obtained here are about 1.8 times higher than in Fig. 2C of the paper, but the curve has the same shape. The original C++ code gives the same level as the Python code, so the difference is between the published code and the published figure. It may come from the way captures are counted. Second, the escape ratio obtained here stops increasing in the ordered phase, while that of the paper keeps increasing. With 40 runs per point it is noisy, so its exact shape is uncertain, and more runs will be needed for the next steps. Finally, the link between inter-individual distance and captures only holds across alignment strengths. At a fixed alignment strength, schools that are more spread out are not caught less. Both quantities may simply depend on the alignment strength, without one causing the other; the confused predator will test this link.

The reproduction uses a single noise level, $D = 0.5$. It also measures the school only once per run, when the predator appears, which may be too early for schools near the transition to settle. The evolution of alignment was not reproduced, as it is outside the scope of this project.

Code, data, figure scripts and the source of this report are available on GitHub.

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